ANAT1009 – Lab #5: The Respiratory System

Purpose:

- To explore the anatomical structures of the respiratory system
- To understand the physiology of ventilation and control mechanisms of breathing.

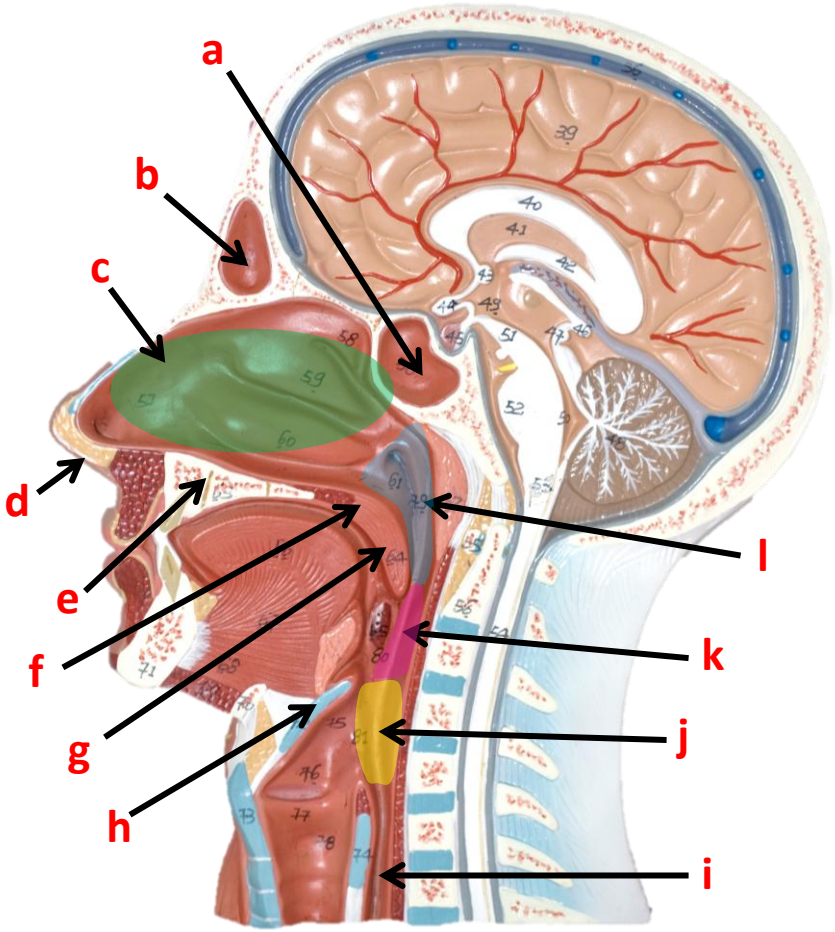
The Appendix, located at the end of the lab, lists the structures for which you are responsible.

Video Available:
Upper respiratory tract structures – Cadaveric prosection

Video Available:
Respiratory System Introduction

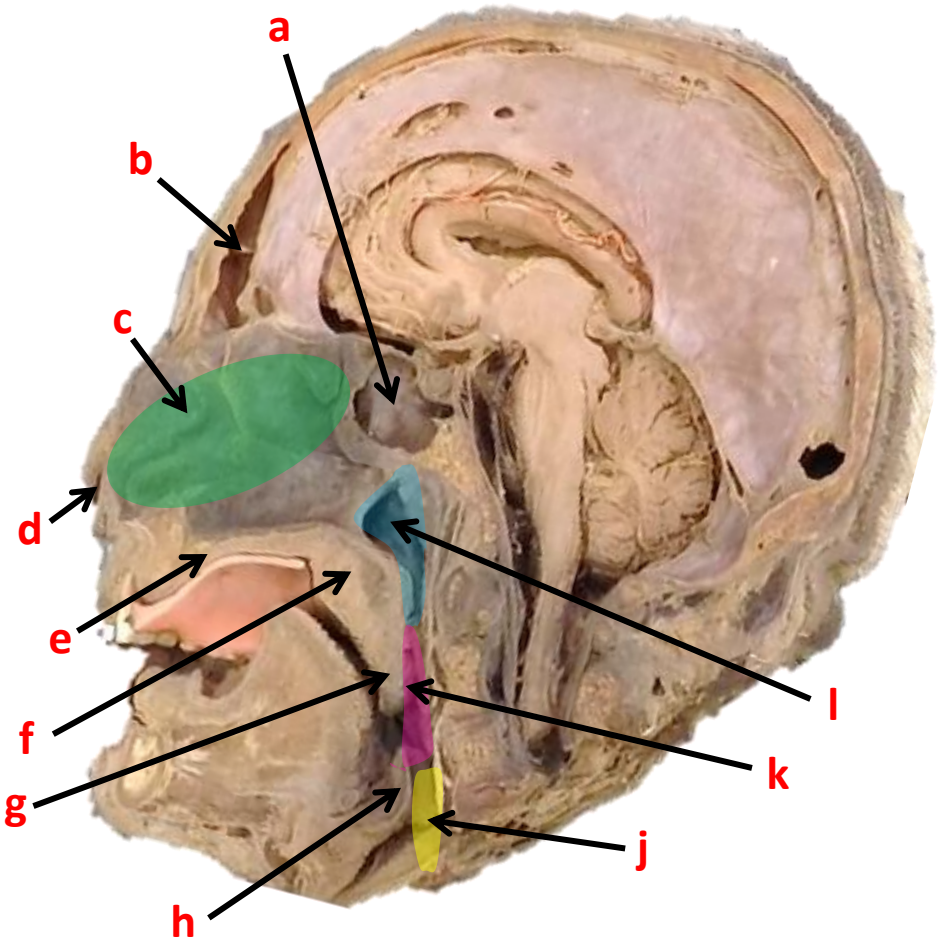
Station 1: Internal structures of the Head and Neck

1. Identify and label the structures listed in the appendix on the images below.



sciencephoto.com

Medial view of sagittal section



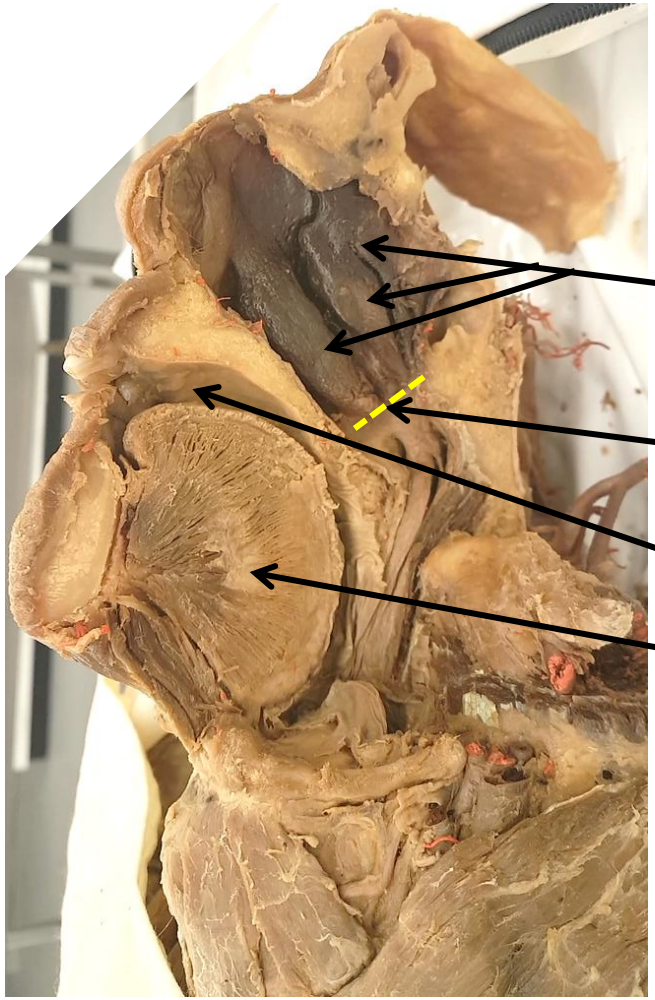
Station 1: Internal Structures of the Head and Neck

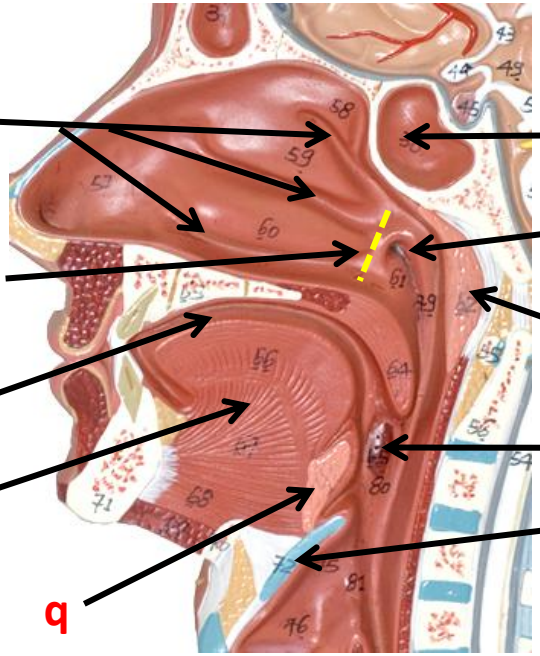
1. Identify and label the structures listed in the appendix on the images below.



Rotational Image Available:
Rotational Images -> Respiratory System -> Sagittal Head and Neck

Video Available:
The upper and lower respiratory





Medial view of sagittal section



m

n

o

p

q

r

s

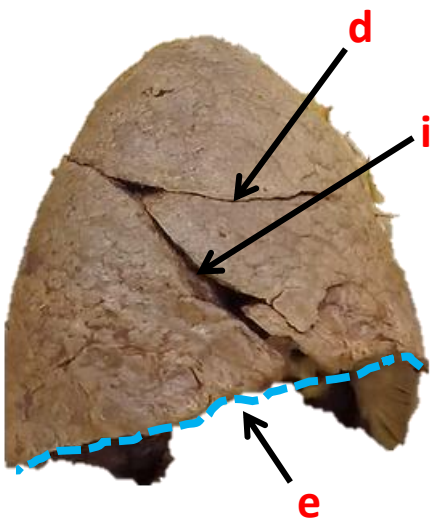
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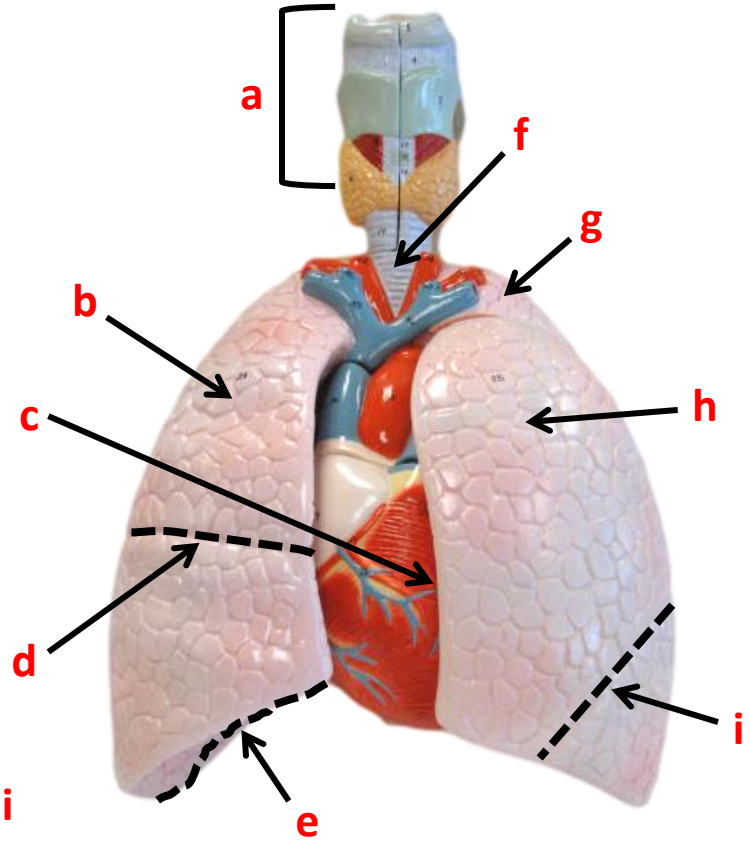
v

Station 2: Structures of the Thorax

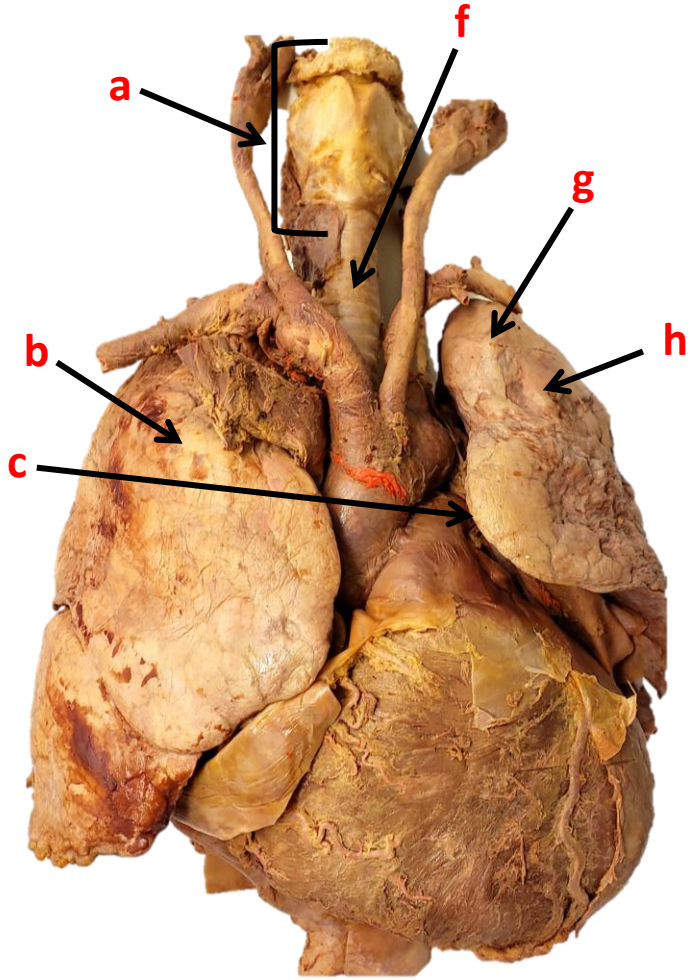
1. Identify and label the structures listed in the appendix on the images below.







Anterior view

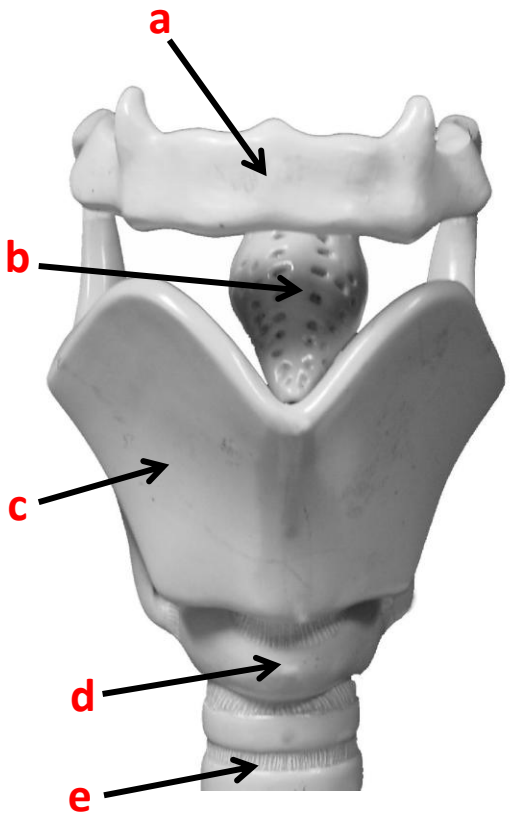


Station 3: The Larynx

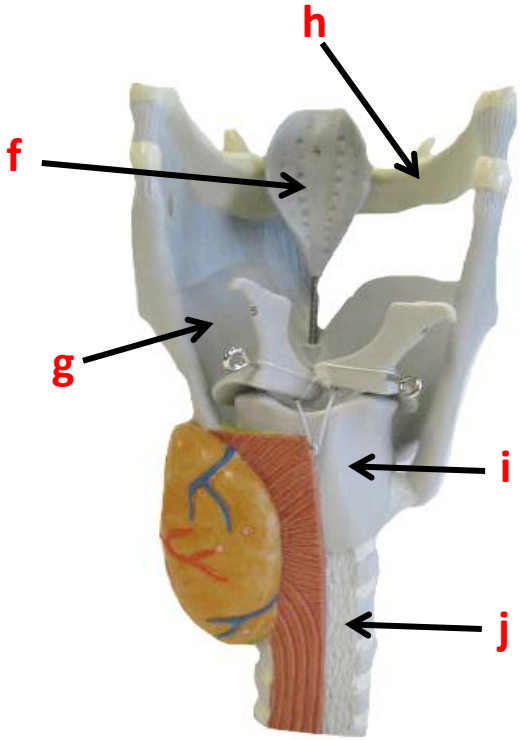
1. Identify and label the structures listed in the appendix on the images below.



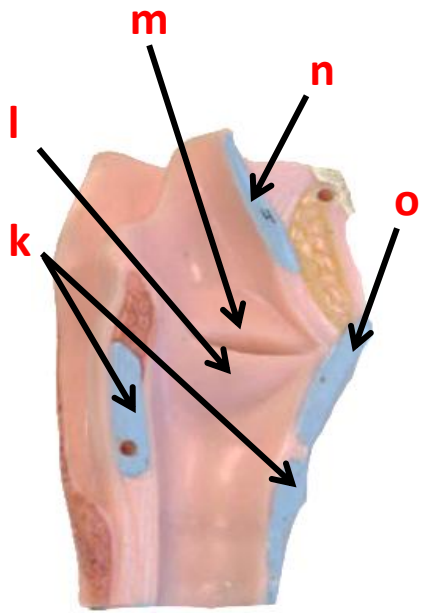
Rotational Image Available:
Rotational Images ->
Respiratory System ->
Respiratory Conducting Zone



Anterior view



Posterior view



Medial view of a sagittal section

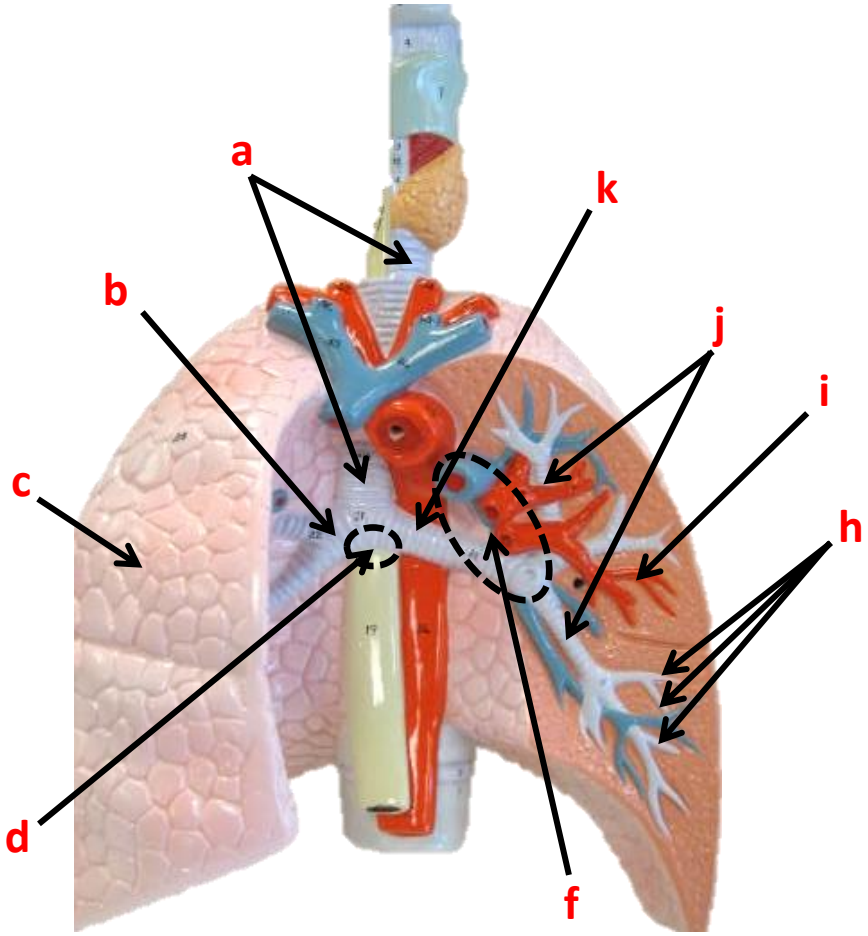
Questions:

1. What is the anatomical name for the "Adam's Apple"?

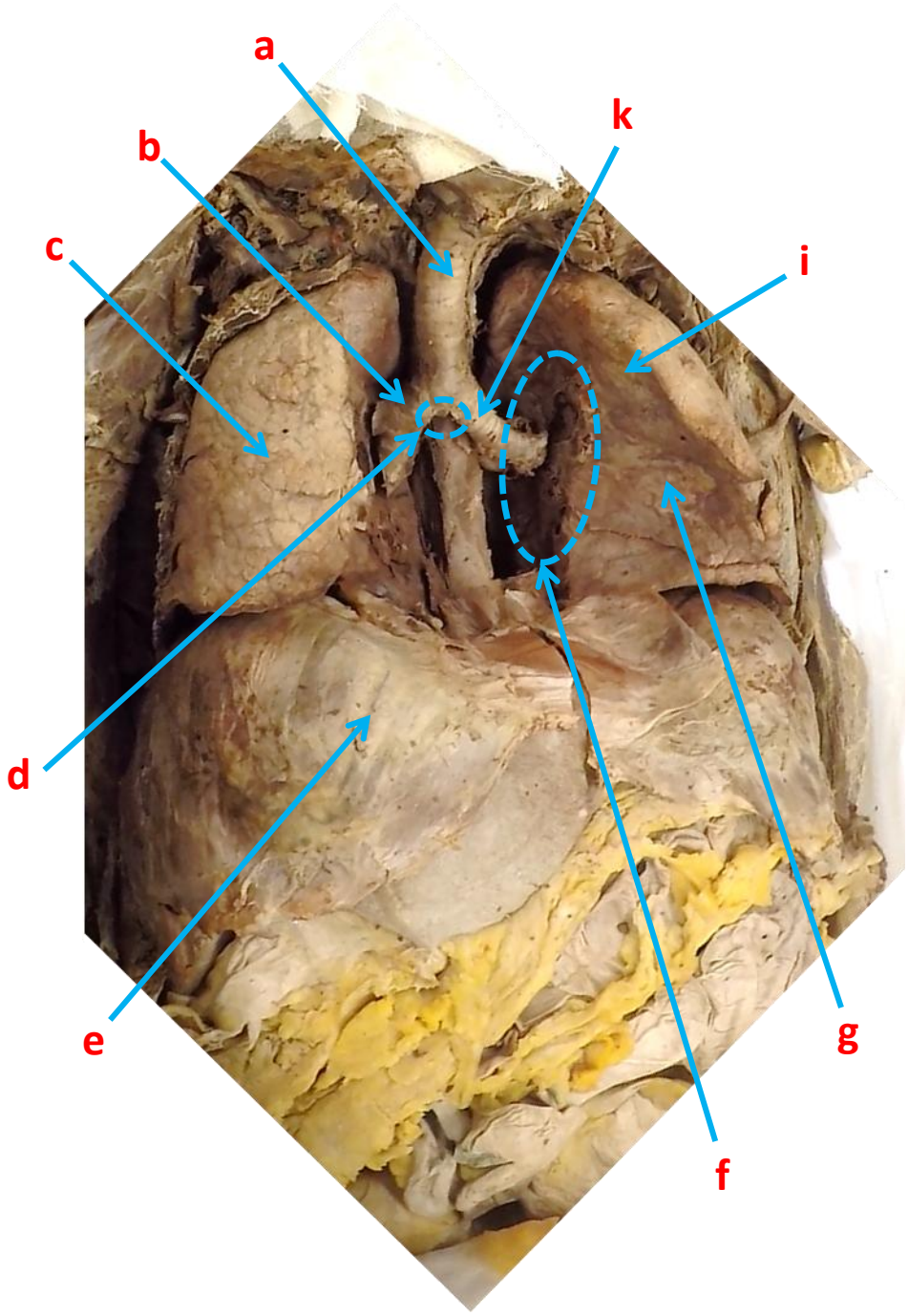
2. What is the difference between the true and false vocal cords?

Station 4: Anatomy of the Lungs

1. Identify and label the structures listed in the appendix on the images below.



Anterior view

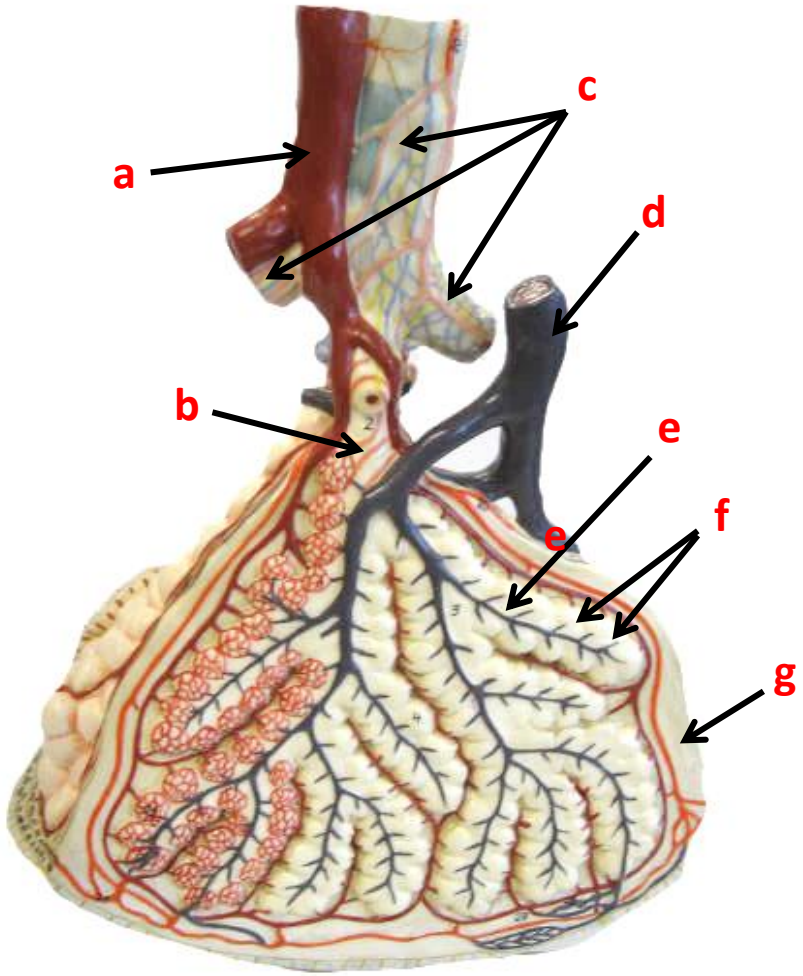


Question:

1. What are the hilus, carina, and cardiac notch?

Station 5: Lung Microstructure

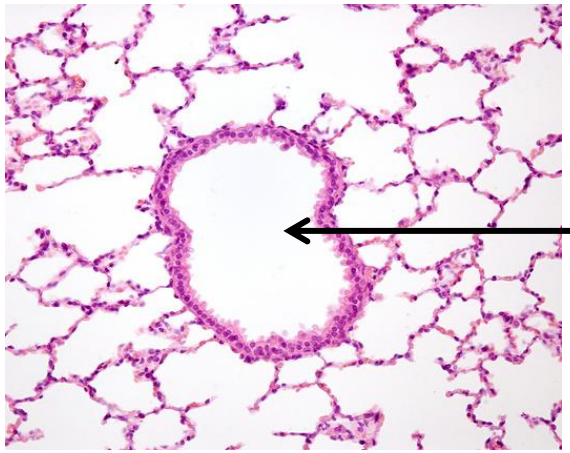
1. Identify and label the structures listed in the appendix on the images below.



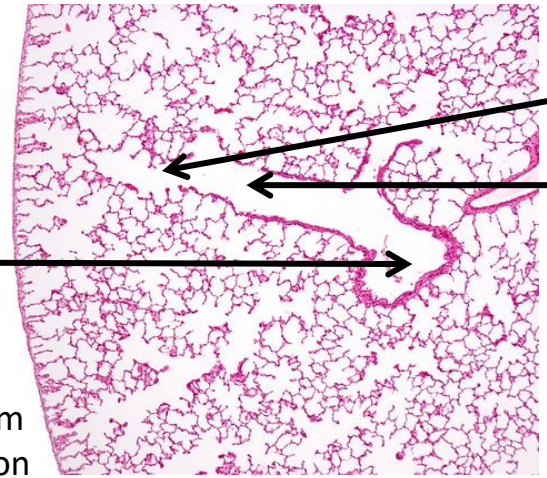
Station 6: Histology of the Lungs

1. Identify and label the structures listed in the appendix on the images below.

1

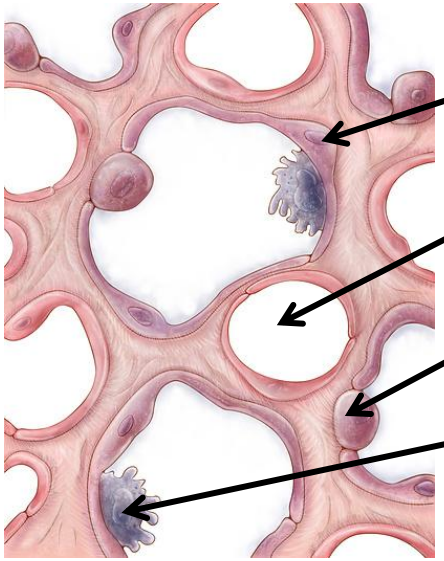


2

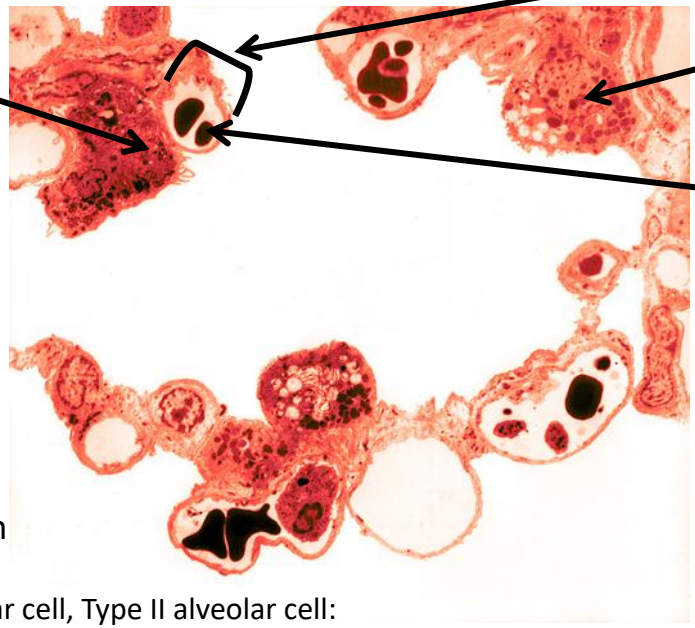


Low/medium magnification

3



4



High magnification

Question:

1. Identify the function of the following cells: macrophage, Type I alveolar cell, Type II alveolar cell:

2. Identify the individual layers and composition of the respiratory membrane:

Station 7: Surfactant, Surface Tension, and Compliance

TO BE DONE AS A DEMO

A: Surfactant and Surface Tension Relationship

Procedure:

1. Take two microscope slides and place them one on top of the other.
2. Repeat step 1 after adding a couple drops of water to the top of one slide.
3. Dry off the slides and repeat step 1 after adding a couple drops of soapy water to the top of one slide.

B: Lung Compliance

Procedure:

1. Observe the demonstration model (two balloons connected via a valve) and identify what the components are supposed to represent.
2. Note the initial relative size of each balloon. Before opening the valve, hypothesize what you would expect to happen in terms of air movement.
3. Open the valve and observe the effects on both balloons.

Questions:

1. What caused the effect you observed when you placed a film of water, vs a film of soapy water between the two slides?

2. What caused the air to move in the direction it did when you opened the valve?

Station 8: Lung Apparatus

Procedure:

1. Observe the model lung, which demonstrates the principles involved as air moves in and out of the lungs. It is a simple device with a bottle “thorax”, a rubber membrane “diaphragm”, and a balloon “lungs”.
2. Move the rubber diaphragm up and down and note the changes in balloon (lung) size as the volume of the thoracic cavity is increased or decreased.
3. Record your observations below and explain physiologically why the lung inflates or deflates when it does.



Station 9: Rebreathing Exercise

Procedure:

1. Have a subject breathe normally and count the number of times the subject breathes, every 15s for 2min. Make sure the subject does not alter their breathing in anyway.
2. Record the results in Table 1 below.
3. Then have the subject breathe into a paper bag. Make sure the paper bag is held tightly around the mouth and nose. Be sure to avoid any leakage of air from the bag.
4. Record ventilation rate for approximately 2 minutes, measuring breathing rate every 15s. Record the results in Table 1 below.

Table 1: Breathing Rate Observations

	Breathing Rate	
Time (sec)	**Normal Breathing**	**Rebreathing in a Paper Bag**
15		
30		
45		
60		
75		
90		
105		
120		

Questions:

1. What happened to the rate and depth of breathing when the subject was breathing into the paper bag? Explain the mechanism involved.

Station 10: Spirometry

Note: A major source of variability in spirometry data originates from the patient so please pay close attention to your breathing maneuvers during the experiment and listen carefully to any instructions you are given.

Procedure:

This test should be performed by only one person in each group. If more time remains once you complete this lab, each person may come back and use the spirometer.

1. Ensure you are standing during testing.
2. Using the available spirometer, determine the various lung volumes indicated in Table. 1

Questions:

1. Define the following and indicate the normal values for each:
    a) VC

    b) IC

    c) ERV

    d) IRV

    e) RV

    f) TLC

Table 1: Spirometry Results – SVC Experiment

	Predicted	1st	2nd	3rd
VC				
IC				
ERV				
TV				

Station 11: Practice Bellringer

In a bellringer, you'll have limited time to answer each question and name the structure shown. You can go through this practice at your own pace, but it's a good idea to try and move through the questions as you would in a test situation.

Use the table below to record your answers.

1.		11.	
2.		12.	
3.		13.	
4.		14.	
5.		15.	
6.		16.	
7.		17.	
8.		18.	
9.		19.	
10.		20.	

Appendix

This list includes the terms you are responsible for knowing. Use it to guide you through the lab. Any of these structures may appear on the bellringer, possibly on different models or specimens than those used in the lab.

If a structure has a left and a right, be sure to include that in your answer.

Station 1: Head and Neck

External nares
Internal nares
Nasal cavity
Nasal concha(e)
Frontal sinus
Sphenoid sinus
Maxillary sinus
Ethmoid sinus*
Hard palate
Soft palate
Uvula
Auditory tube
Oral cavity
Tongue
Epiglottis
Esophagus
Trachea
Nasopharynx
Oropharynx
Laryngopharynx
Pharyngeal tonsil
Lingual tonsil
Palatine tonsil

Station 2: The Thorax

Larynx
Trachea
R/L lung
Base of lung
Apex of lung
Cardiac notch
Horizontal fissure
Oblique fissure

Station 3: The Larynx

Epiglottis
Hyoid bone
Thyroid cartilage
Cricoid cartilage
Trachea
True vocal cords/vocal folds
False vocal cords/vestibular folds

Station 4: Anatomy of the Lungs

R/L lung
Trachea
Carina
R/L main/primary bronchus
R/L lobar/secondary bronchus
R/L segmental/tertiary bronchus
Hilus/hilum
Cardiac notch
Diaphragm

Station 5: Lung Microstructure

Pulmonary vein
Pulmonary artery
Terminal bronchiole
Respiratory bronchiole
Alveolar duct
Alveoli
Visceral pleura

Station 6: Lung Histology

Terminal bronchiole
Respiratory bronchiole
Alveolar duct
Type I alveolar cell
Type II alveolar cell
Alveolar macrophage
Capillary
Red blood cell

***Be aware when you pluralize a word it MAY change the spelling of the word. e.g., bronchus becomes bronchi**

***Term is not included in this lab handout but you are responsible for this term.**